

Lab Report 6 : Creating Stopwatch App

Course Title: Mobile Application Development

Course Code: CSE-410



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GitHub Repository:

You can find the repository at the following link:

<https://github.com/RubayedMunna/StopWatchApp.git>

Code:

activity_stopwatch.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    tools:context=".StopwatchActivity">

    <TextView
        android:id="@+id/time_view"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="center_horizontal"
        android:textAppearance="@android:style/TextAppearance.
            Large"
        android:textSize="56sp" />

    <Button
        android:id="@+id/start_button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="center_horizontal"
        android:layout_marginTop="20dp"
        android:onClick="onClickStart"
        android:text="@string/start" />

    <Button
        android:id="@+id/stop_button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="center_horizontal"
        android:layout_marginTop="8dp"
        android:onClick="onClickStop"
        android:text="@string/stop" />

    <Button
        android:id="@+id/reset_button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```
        android:layout_gravity="center_horizontal"
        android:layout_marginTop="8dp"
        android:onClick="onClickReset"
        android:text="@string/reset" />
</LinearLayout>
```

Listing 1: activity_stopwatch.xml

StopwatchActivity.java

```
package com.example.stopwatchapp;

import android.app.Activity;
import android.os.Bundle;
import android.view.View;
import java.util.Locale;
import android.os.Handler;
import android.widget.TextView;
import android.widget.Toast;
import com.google.android.material.snackbar.Snackbar;

public class StopwatchActivity extends Activity {
    //Number of seconds displayed on the stopwatch.
    private int seconds = 0;
    //Is the stopwatch running?
    private boolean running;
    private boolean wasRunning;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_stopwatch);
        if (savedInstanceState != null) {
            seconds = savedInstanceState.getInt("seconds");
            running = savedInstanceState.getBoolean("running");
            wasRunning = savedInstanceState.getBoolean("wasRunning");
        }
        runTimer();
    }

    @Override
    public void onSaveInstanceState(Bundle savedInstanceState) {
        savedInstanceState.putInt("seconds", seconds);
        savedInstanceState.putBoolean("running", running);
        savedInstanceState.putBoolean("wasRunning", wasRunning);
    }

    @Override
    protected void onPause() {
        super.onPause();
        wasRunning = running;
        running = false;
    }

    @Override
```

```
protected void onResume() {
    super.onResume();
    if (wasRunning) {
        running = true;
    }
}

//Start the stopwatch running when the Start button is clicked.
public void onClickStart(View view) {
    running = true;

    // Show a toast indicating the stopwatch has started
    Toast.makeText(this, "Stopwatch started", Toast.LENGTH_SHORT).
show();
}

//Stop the stopwatch running when the Stop button is clicked.
public void onClickStop(View view) {
    running = false;

    // Calculate the current time (hours, minutes, seconds)
    int hours = seconds / 3600;
    int minutes = (seconds % 3600) / 60;
    int secs = seconds % 60;

    // Format the time as a string
    String currentTime = String.format(Locale.getDefault(), "%d:%02
d:%02d", hours, minutes, secs);

    // Show a Snackbar that remains visible
    Snackbar.make(view, "Stopwatch stopped at " + currentTime,
Snackbar.LENGTH_INDEFINITE)
        .setAction("Dismiss", new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                // This will dismiss the Snackbar when any
action is performed
            }
        }).show();
}

//Reset the stopwatch when the Reset button is clicked.
public void onClickReset(View view) {
    running = false;
    seconds = 0;

    // Show a toast indicating the stopwatch has been reset
    Toast.makeText(this, "Stopwatch reset to 0:00:00", Toast.
LENGTH_SHORT).show();
}

//Sets the number of seconds on the timer.
private void runTimer() {
    final TextView timeView = (TextView) findViewById(R.id.
time_view);
    final Handler handler = new Handler();
    handler.post(new Runnable() {
```

```
@Override
public void run() {
    int hours = seconds / 3600;
    int minutes = (seconds % 3600) / 60;
    int secs = seconds % 60;
    String time = String.format(Locale.getDefault(),
        "%d:%02d:%02d", hours, minutes, secs);
    timeView.setText(time);
    if (running) {
        seconds++;
    }
    handler.postDelayed(this, 1000);
}
});
}
```

Listing 2: StopwatchActivity.java

strings.xml

```
<resources>
    <string name="app_name">StopWatchApp</string>

    <string name="start">Start</string>
    <string name="stop">Stop</string>
    <string name="reset">Reset</string>
</resources>
```

Listing 3: strings.xml

Output:

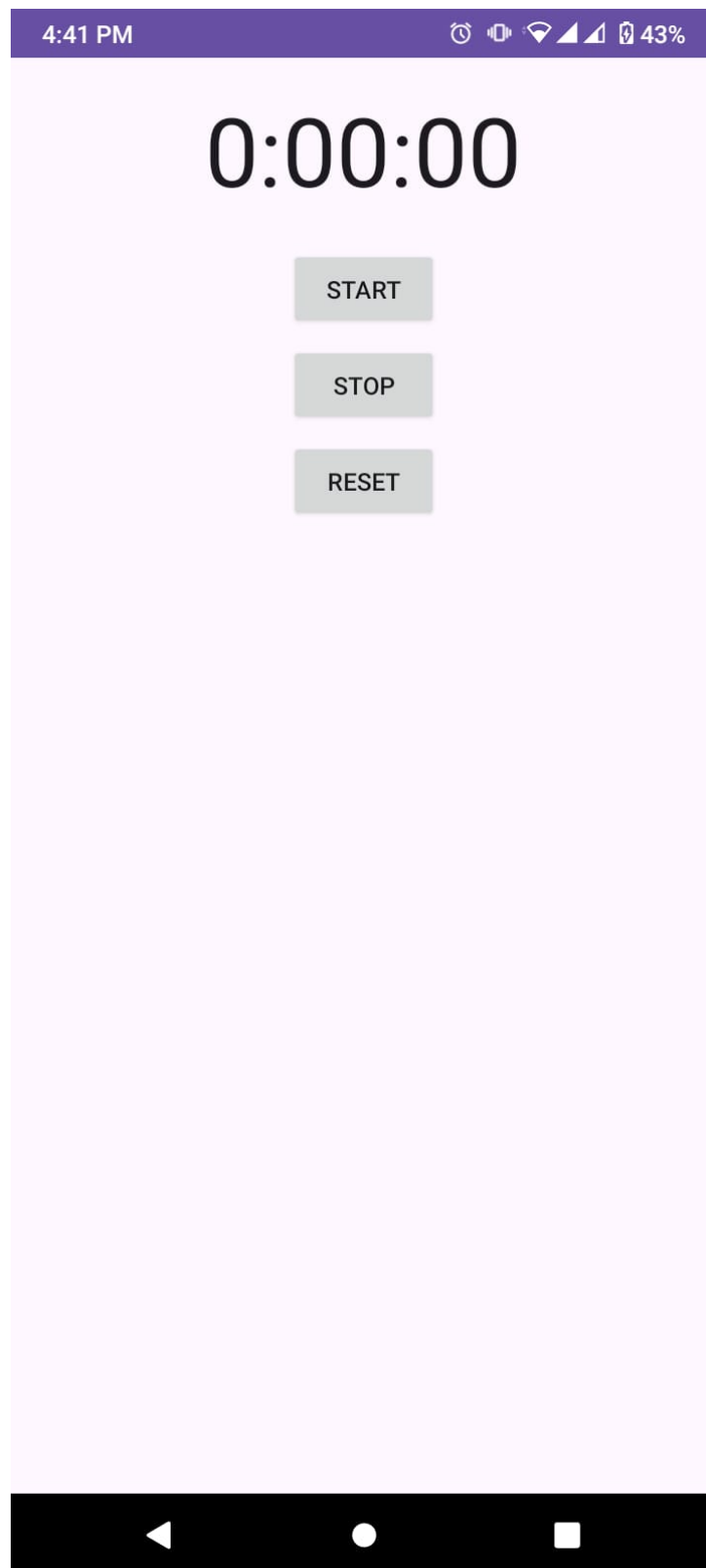


Figure 1: First Look of The App

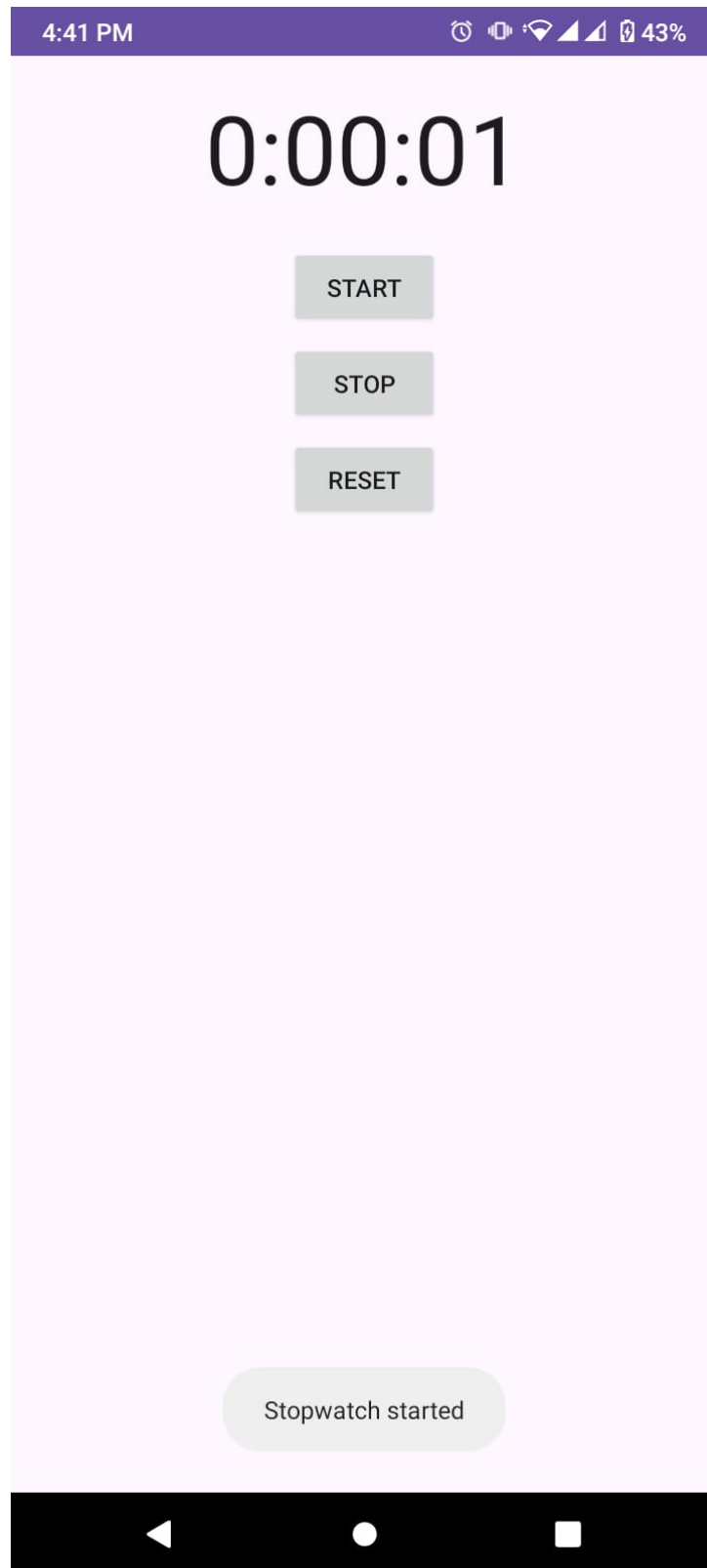


Figure 2: Stopwatch Started

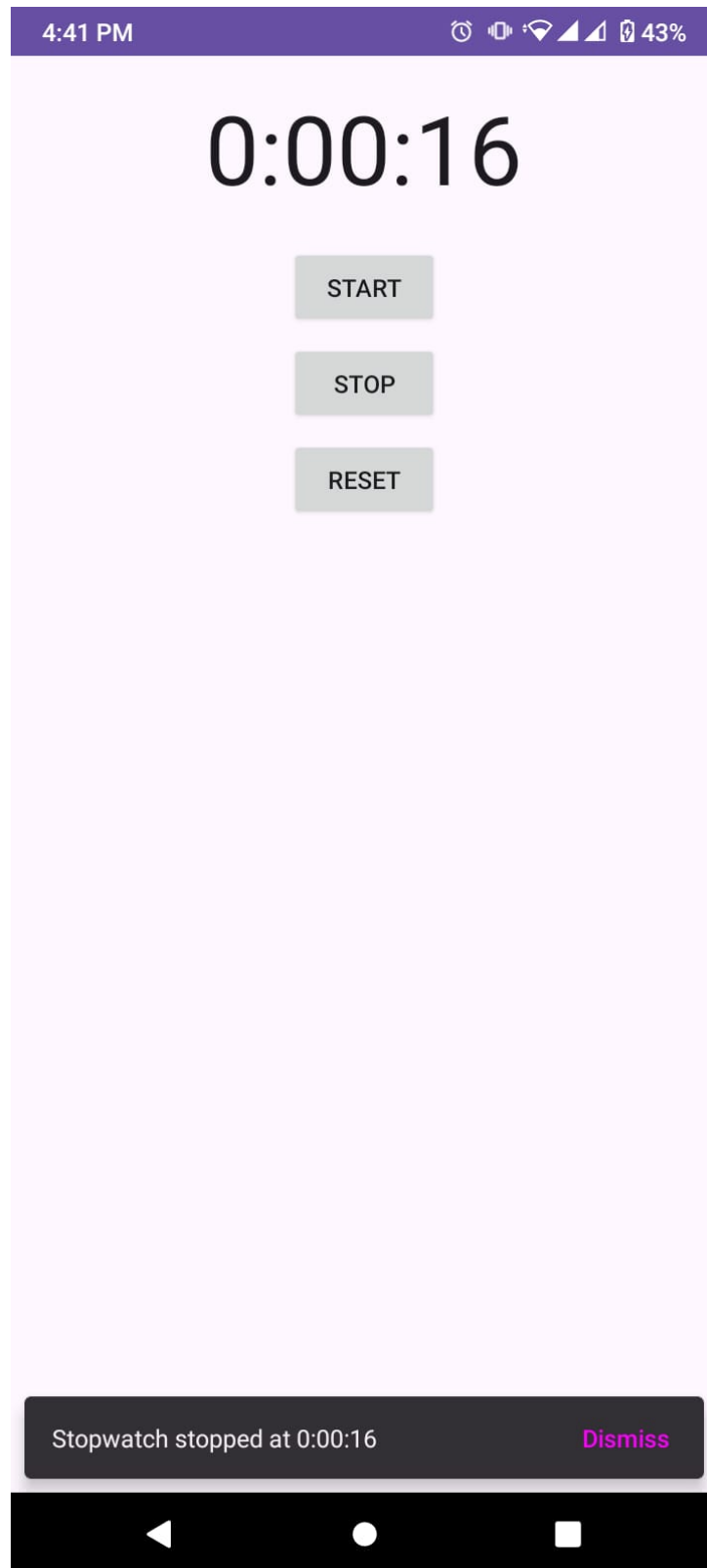


Figure 3: Stop Button Is Clicked

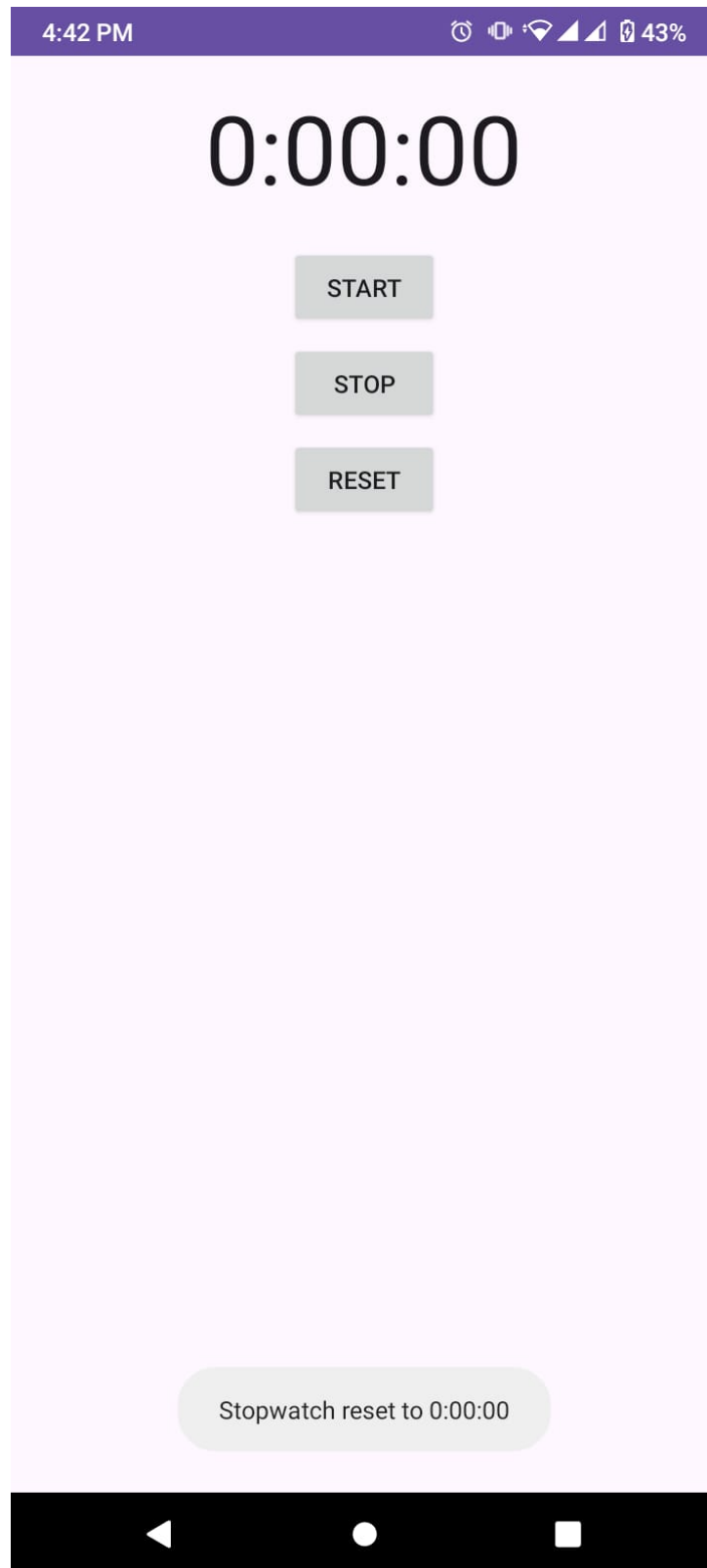


Figure 4: Stopwatch is reset to zero

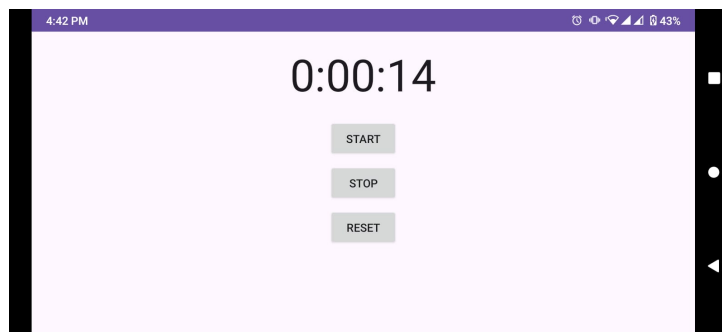


Figure 5: Horizontal orientation of stopwatch